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Effects of Living Kidney Donation on Arterial Stiffness: A Systematic Review Protocol

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Effects of Living Kidney Donation on Arterial Stiffness:

A Systematic Review Protocol

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Abstract

Introduction: Losing renal mass to unilateral nephrectomy in kidney donors has been associated with an accelerated progression of aortic stiffness and decreased glomerular filtration compared with healthy non-donors. This is a concern because increased aortic stiffness is an independent predictor of overall cardiovascular disease and all-cause mortality in the general population. To confirm if arterial stiffness increases after donation, we will systematically review all studies that evaluated indices of arterial stiffness in healthy individuals who underwent unilateral nephrectomy for kidney donation compared to age-matched healthy non-nephrectomized controls.

Methods/Analysis: We will comprehensively search for studies published after January 1, 1960 in MEDLINE, EMBASE, Cochrane Central, OVID, and EBM reviews. All prospective (cohort, case-control, case series, and before-and-after studies) and retrospective non-randomized studies reporting indices of arterial stiffness in nephrectomized and non-nephrectomized healthy participants will be included. Primary outcome will be the difference in the functional metrics of arterial stiffness between donors and non-donors. Secondary outcomes will be the differences in systolic/diastolic blood pressures, serum creatinine, glomerular filtration, intima-media thickness and vascular calcification. Study screening, selection and data extraction will be performed by 2 independent reviewers. Risk of bias will be independently assessed with the ROBINS-I tool and confidence in evidence by the GRADE recommendations. Qualitative and quantitative data syntheses as well as clinical and statistical heterogeneity (Forrest plots, I^2 and Cochran's Q statistics) will be evaluated. If clinical and statistical heterogeneity are acceptable, inverse variance-weighted effects will be analyzed by random effect models.

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Ethics and dissemination: No ethical approval is necessary. Our results will be disseminated through peer-review publication and presentations to guide stakeholders on the evaluation and follow-up care of kidney donors.

PROSPERO registration number: CRD42020185551.

Key words: vascular, arterial, aortic stiffness, organ donation, kidney donor, cardiovascular disease, pulse wave velocity.

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Strengths and limitations of this study

- This systematic review will synthesize best quality evidence on functional and structural components of arterial stiffness in kidney donors and age-matched healthy non-nephrectomized controls.
- Since systemic blood pressure may be a determinant of arterial stiffness, its inclusion in a secondary analysis will improve our understanding on the potential relationship between blood pressure and arterial stiffness after donation.
- Quality of evidence and differences in study design can limit the strength of our associations. To minimize misinterpretations, our study will implement well-validated scales for assessing risk of bias and confidence in cumulative evidence.
- The results of this review can highlight strengths and weaknesses in previous studies and determine if there is a need for further well-designed investigations in kidney donors.

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INTRODUCTION

Live donor kidney transplantation is the preferred treatment for patients with end-stage renal disease (ESRD); it provides increased patient and graft survival and better quality of life than either dialysis or deceased donor transplantation.[1,2] Every year, nearly 20,000 healthy adults worldwide accept the risk of donor nephrectomy to help family members, friends or even strangers to improve survival and quality of life.[3,4] From a medical and ethical perspective, we ought to minimize their risks and maximize safety during and after kidney donation.

Although several studies [2,5,6] have indicated that the absolute risk for healthy kidney donors following nephrectomy is very low, some have reported that unilateral nephrectomy is associated with an accelerated progression of aortic stiffness, enhanced left ventricular mass, and decreased glomerular filtration compared with healthy non-donors matched by age and sex.[7-10] In most of these reports, blood pressure at follow-up was not significantly different between donors and non-donors.[8,9,11,12] Moreover, the risk of developing hypertension after donation was associated with higher blood pressure readings before donation, suggesting that reduced glomerular filtration rate (GFR) post-donation may be a graded risk factor for increased aortic stiffness, independent of blood pressure.[7,12] These findings are clinically relevant because increased aortic stiffness as measured by carotid-femoral pulse wave velocity (cf-PWV) is a strong independent predictor of overall cardiovascular disease and all-cause mortality in the general population.[13] An increase of 1 m/s in cf-PWV in an otherwise healthy individual has been associated with an adjusted risk increase of 14% and 15% in total cardiovascular events and all-cause mortality.[13] Although compensatory renal hypertrophy and glomerular hyperfiltration occur after unilateral nephrectomy, a net decrease in total GFR typically follows, and this raises the possibility that cardiovascular disease risk after donation may become comparable to the risk in patients with mild

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to moderate chronic kidney disease.[14-16] A reduction in renal function of such magnitude may be particularly critical for young living kidney donors who have the longest duration of risk exposure to the effects of reduced renal mass. To determine if kidney donors are at risk for increased aortic stiffness, a systematic appraisal of existing studies on arterial stiffness in donors compared to age-matched healthy controls is necessary. This review differs from existing systematic reviews and meta-analyses that have evaluated the interventional effects of kidney transplantation on arterial stiffness in kidney recipients with ESRD.[17,18] The results of this new review can highlight strengths and weaknesses in previous studies with kidney donors and determine if there is a need for further well-designed investigations.

Objectives

In this study, we will systematically review all non-randomized studies that evaluated central or peripheral arterial stiffness in healthy individuals who underwent unilateral nephrectomy after fulfilling standard criteria for kidney donation and who had measurements of the functional or structural components of arterial stiffness, as defined by validated metrics including: carotid-femoral PWV, carotid-radial PWV, brachial-ankle PWV, pulse wave analysis, femoral-tibial PWV, ankle-brachial index, aortic distensibility, carotid-intima media thickness and vascular calcification. The major objective of this review will be to provide best quality evidence on whether arterial stiffness increases after living kidney donation beyond what is expected for age-matched healthy non-nephrectomized controls. We will primarily determine differences in functional markers of arterial stiffness between kidney donors and their respective comparative healthy non-donor controls. Secondarily, we will document differences in systolic and diastolic blood pressure (SBP, DBP), serum creatinine, estimated GFR (eGFR) and structural markers of

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arterial stiffness (carotid-intima media thickness, arterial calcification scores) between these 2 populations.

METHODS AND ANALYSIS

We will conduct this review in accordance with the Cochrane Collaboration principles for Systematic Reviews[19] and this protocol has been registered in the PROSPERO register (CRD42020185551) for systematic reviews.[20] The preferred reporting items for Systematic Reviews described by the PRISMA guidelines[21] will be followed and a checklist file for these recommendations will be included (Supplementary appendix 1).[22]

Population and eligibility criteria

Kidney donors

We will include healthy adults (≥ 18 years old) of any age who underwent unilateral nephrectomy after meeting standard screening criteria for organ donation and whose arterial stiffness was measured non-invasively before and/or after nephrectomy with any of the following functional metrics of arterial stiffness: carotid-femoral PWV, carotid-radial PWV, brachial-ankle PWV, femoral-tibial pulse PWV, ankle-brachial index, pulse wave analysis (augmentation index and central pulse pressure) and aortic distensibility.[23] As indicators of structural change in vascular stiffness, measurements of carotid-intima media thickness and arterial calcification scores will also be documented.[24]

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Healthy controls

As controls, we will include adult individuals (>18 years of age) with measurements of validated indices of arterial stiffness who participated as healthy comparative controls in kidney donor studies.[25]

Intervention

In this review, uneventful unilateral nephrectomy for kidney donation independent of the surgical approach (laparoscopic and open nephrectomy) will be considered as the main intervention.

Outcomes

The primary outcome will be the difference in the functional metrics of arterial stiffness between healthy nephrectomized and non-nephrectomized participants. We will report absolute values as well as between-group mean differences in their respective units of measurement. Secondary outcomes will be the differences in systolic and diastolic blood pressures (mm Hg), pulse pressure (mm Hg), serum creatinine levels (μmol/L), eGFR (ml/min/1.73 m^2), intima media thickness (mm) and vascular calcification (scores). Clinical outcomes and any additional measures of endothelial function will be documented and reported accordingly.

cf-PWV reference group

In addition to the primary comparative analysis, we will search for studies that measured cf-PWV in different healthy populations worldwide and who have reported values of normal cf-PWV standardized by age.[26] We expect that this analysis will improve the generalizability of this review.

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Study design

Since prospective randomized clinical trials (RCTs) of donation would never be possible for ethical reasons, only prospective non-randomized (cohort, case-control, case series, and before-and-after studies) and retrospective studies will be included, provided that 10 or more subjects have participated in the primary analysis.

Search strategy

Our search strategy will be conducted using MEDLINE, EMBASE, Cochrane Central databases, CADTH's "Grey matters light", OVID, EBM reviews, Health Technology Assessment (HTA) database and other grey literature of studies including government reports and policy statements published after January 1, 1960. A comprehensive search strategy will be constructed and implemented by a health information specialist (RS) with systematic review experience, in collaboration with the research team. MeSH terms will be used to capture the principal elements of the research question and identified citations will be exported to an excel spreadsheet for study review and selection. Our proposed literature search strategy is outlined in the supplementary appendix 2. Manual review of the reference lists for all studies will be conducted according to pre-defined screening criteria (Table 1). A final "grey literature" search will be conducted using "Google Scholar" as well as reviews of *clinicaltrials.gov* for any current study. No language restriction will be utilized in any of the initial search, but our final analysis will be restricted to articles reported in English, French, Italian, Portuguese and Spanish languages. Duplicate citations will be removed and search strategies will be kept up to the time of the end of this review.

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Table 1. List of inclusion and exclusion criteria for the selection of comparative groups.

	Inclusion criteria	Exclusion criteria
Population	• Healthy subjects (≥ 18 years) who underwent unilateral simple nephrectomy as selected candidates that met standard criteria for kidney donation	• Healthy participants that underwent unilateral simple nephrectomy other than for kidney donation. • Children and adolescents with solitary kidney after unilateral nephrectomy
Intervention	• Open or laparoscopic unilateral simple nephrectomy	
Comparator	• Healthy adult subjects (>18 years) with measurements of validated indices of arterial stiffness who participated as healthy comparative controls in kidney donor studies • Healthy subjects with measurements of carotid-femoral PWV included in reference studies	• Kidney recipients
Outcome	• Functional indices of arterial stiffness: carotid-femoral pulse wave velocity (PWV), carotid-radial PWV, brachial-ankle PWV, femoral-tibial PWV, pulse wave analysis (augmentation index and central pulse pressure), ankle-brachial index, aortic distensibility, • Structural indices of arterial stiffness: carotid-intima media thickness and arterial calcification scores.	• None
Study design	• Prospective non-randomized (cohort, case-control, case series, and before-and-after studies) and retrospective studies if 10 or more participants have been included in the primary analysis • Articles reported in English, French, Italian, Portuguese and Spanish languages	• Pediatric and non-human studies • Narrative reviews • In-vitro or mathematical modeling reports • Duplicates • Sub-studies of previously published trials. • Abstracts, conference proceedings

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Study screening, exclusions and selection

During the screening phase, we will include all prospective and retrospective non-randomized studies reporting measurements of arterial stiffness in nephrectomized and non-nephrectomized healthy participants. An iterative process of study selection will be followed using the inclusion and exclusion criteria set out in Table 1. We will exclude studies reporting on investigations in pediatric populations (age < 18 years), non-human studies, narrative reviews, in-vitro or mathematical modeling reports, abstracts or conference proceedings and any duplicate or sub-study of previously published investigations. All review processes will be independently performed by 2 individuals, with a third person available for consensus in cases of discrepancies.

All citations will first be screened by title and abstract. After selection by title/abstract and with the purpose of facilitating a final decision on eligible studies, an intermediate screening of the material and methods section will be implemented in duplicate to confirm the type of methodology employed for measuring arterial stiffness. A judicious decisive selection before data extraction will be achieved by the independent reviewers at the full-text level. In cases of missing information, we will consider contacting study authors. Neither of the review authors will be blinded to the journal titles or to the study authors or institutions.

Data extraction

A data extraction form will be prepared a-priori with consensus from all investigators and optimized using a subset of 4 randomly selected studies prior to duplicate extraction by 2 independent reviewers. When multiple publications arise from one study, relevant data will be extracted into a single form. Data extraction will include: a) *study characteristics, design and methods*: title, authors, journal/source/year, language of publication, country, type of study design,

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study period, publication status, total number of donors and non-donors, inclusion and exclusion criteria, and points of measurement; b) *sample characteristics*: age, sex, age at time of nephrectomy, age at time of assessment, duration of follow-up and type of arterial stiffness instrumentation; c) *outcomes*: change in arterial stiffness based on the metric of measurement: carotid-femoral PWV, carotid-radial PWV, brachial-ankle PWV, femoral-tibial pulse PWV, ankle-brachial index, augmentation index, central pulse pressure, aortic distensibility, carotid artery intima-media thickness; arterial calcification scores, serum creatinine levels, eGFR, systolic, diastolic and pulse blood pressure. If reported, we will document type and number of medications for the control of blood pressure.

Risk assessment of bias

Risk of bias for non-randomized studies will be assessed using the ROBINS-I tool.[27] Seven domains through which bias might be introduced will be independently evaluated and discussed by 2 reviewers, encompassing confounding, selection of participants (nephrectomized and non-nephrectomized), classification of the intervention (i.e. unilateral nephrectomy), deviation from the intended intervention, missing data, measurement of outcomes and selection of the reported results. Each domain will be judged as either low, moderate, serious or critical risk of bias or no information available. After discussion among reviewers, an overall assessment of study bias with inclusion of all domains will be tabulated. Unresolved disagreements will be resolved with participation of a third reviewer.

Data synthesis and analysis plan

Studies will be included in qualitative and quantitative syntheses if they fulfill all eligibility criteria. Study characteristics will be summarized using means and standard deviations or median

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and inter-quartile ranges for continuous variables and numbers and percentages for categorical variables. A narrative report of study characteristics will also be provided. We will identify potential sources of clinical heterogeneity according to differences in study design characteristics, methodological quality, characteristics at baseline between nephrectomized groups and their controls, and duration of follow-up periods. If at least 2 studies report on the same outcome, a quantitative analysis (i.e. meta-analysis) will be attempted on those studies. Statistical heterogeneity will be characterized with Forest plots, the I^2 and Cochran's Q statistics ($p<0.05$). If both clinical and statistical heterogeneity are acceptable, inverse variance-weighted effects meta-analysis will be performed. An I^2 value less than 50% with a non-significant chi-square statistic ($P<0.10$) will be suggestive of moderate statistical heterogeneity. We will primarily choose the random effects model according to the methodology of DerSimonian and Laird,[28] but a fixed-effects meta-analysis will also be modeled as part of our sensitivity analysis. Based on the scale of measurements, we may choose to calculate pooled effect estimates using either standardized or mean differences and their corresponding 95% confidence intervals (95% CI). To minimize the "double counting" error in before-and-after studies, half of the total number of study participants will be allocated to each study arm. All analysis will be performed using RevMan 5.3 (The Nordic Cochrane Centre, The Cochrane Collaboration, 2014, Copenhagen).

Sensitivity analysis

Depending on the number of eligible studies, sensitivity analyses may be performed to evaluate the effects of extreme values or high risk of study bias. An analysis of the differences between fixed and random effects models on the pooled effect estimates will also be assessed.

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Subgroup analysis

Clinical studies[29] have documented that living donors genetically related to their ESRD recipients are more likely to develop the primary kidney disease. Thus, it is critical to differentiate the effects of unilateral nephrectomy on arterial stiffness between recipient-related healthy controls and those who are not related to the recipient. To minimize bias in our review and depending on the number of eligible studies and nature of the reported information, we will attempt to determine the effects of potential confounders by performing sub-group analyses based on age, type of comparator (recipient-related versus non-recipient-related), number and type of medications used for the control of blood pressure and sex according to different male to female ratios. Inter-group differences will be analyzed using the Cochran's Q statistics with $p \leq 0.10$.

Confidence in cumulative evidence

To assess the certainty in the evidence and strength of recommendations on the effects of unilateral nephrectomy in kidney donors, 2 reviewers will evaluate quality of evidence for each outcome measure according to the 5 domains of GRADE recommendations.[30]

Amendments

Any protocol amendments will be summarized in the form of a Table, where date of amendment, description of changes and rationale will be provided.

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DISCUSSION

Because living kidney donors are selected amongst the healthiest individuals, the long-term risk of ESRD, cardiovascular disease or death has been reported to be similar to or even lower than that in the general population.[6] It has been documented however, that unilateral nephrectomy frequently leads to reduced kidney function and variable levels of increased serum creatinine.[5,14,15] Although the medium- and long-term adverse effects of hyperfiltration and compensatory hypertrophy after donation remain unclear, endothelial dysfunction associated with decreased brachial artery reactivity[16,31] and increased levels of uremic toxins[32] have been reported in kidney donors. More recently, some studies have reported that unilateral nephrectomy leads to an increase in aortic stiffness, left ventricle mass and decreased GFR as early as one year after surgery.[7-9] Although blood pressure was not markedly affected in these studies, initial results suggest that reduced GFR may be an independent graded risk factor for increased aortic stiffness and adverse left ventricle remodeling, independent of blood pressure.[8,11]

To characterize the hemodynamic effects of unilateral nephrectomy in kidney donors, we propose to systematically identify, appraise and summarize all non-randomized studies that have investigated indices of arterial stiffness before and after nephrectomy in donors to determine any increase in the progression of arterial stiffness compared with age-matched healthy individuals. In addition to improving our understanding of the relationship between reduced kidney function after donation and cardiovascular function, our findings will determine if unilateral nephrectomy is a risk factor for increased arterial stiffness in donors and whether non-invasive metrics of arterial stiffness could be useful for risk stratification.

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Animal studies have shown that alterations in renal development resulting in reduced nephron number in the offspring result in blood pressure elevation, reduced cardiac functional reserve and left ventricle hypertrophy in adulthood.[33] Although alterations of the renin-angiotensin-aldosterone system and changes in vascular tone are frequently present after unilateral nephrectomy,[34] kidney donation is not considered a risk factor for hypertension post-donation. Kasiske et al[35] followed living kidney donors for a 3-year period and identified an increase in blood pressure over time, but differences between donors and controls were non-significant. Kim et al[36] found that there was an increase in blood pressure after donation with 21% of donors requiring medication for control of hypertension. Thiel et al[12] reported that the risk of developing hypertension (defined as >140/90 mm Hg) increased 3.6-fold 1-year post-donation, but baseline values of kidney donors who became hypertensive were significantly higher than those who were less susceptible to hypertension. Because systemic blood pressure may be a determinant of arterial stiffness, we will include systolic and diastolic blood pressure as a secondary outcome. This will add to the understanding of the potential relationship between blood pressure and arterial stiffness following organ donation.[34-36]

To accept that increased arterial stiffness is associated with the effects of unilateral nephrectomy, convincing evidence based on good quality studies and adequate methodological designs is critical. Although quality of evidence and differences in study design are factors that may limit the strength of our associations, implementing well-validated scales in our study for the assessment of risk of bias and certainty of evidence will minimize misinterpretations. In cases where information is not available, we will make every possible effort to contact the primary author(s) to obtain any missing information.

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Conclusions

Living kidney donors obtain no medical benefit from donation, exposing themselves to health risks associated with the surgical procedure and the effects of reduced renal mass for the benefit of another human being. Promotion of donor autonomy in the medical decision-making process requires accurate quantification of the mid and long-term risks after donation. This is particularly critical for young living donors who have the longest risk of exposure.[37,38] Our review will provide a more accurate estimate of the long-term risks of cardiovascular disease after donation by examining the progression of arterial stiffness post-nephrectomy.[34-38] Such results can inform potential donors and may guide their follow-up care and preserve good health in the long-term.

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Ethics and dissemination

No ethics approval is required for this systematic review. Our findings will be disseminated through peer-review publication and presentation, which can inform clinical guidelines and management of potential kidney donors.

Contributors: RAR and KDB conceived and designed the study; RAR created the analytical plan and drafted the protocol; RAR and RS designed the search strategy; RAR, MTS and KDB will be involved in one or more of the following tasks: study screening and selection, data extraction, verification, quality appraisal, synthesis, analysis of the evidence and data interpretation. All authors have read, reviewed and approved the final version of the manuscript. RAR is the guarantor.

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Supplementary Appendix 1: PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

TITLE: **Effects of Living Kidney Donation on Arterial Stiffness: A Systematic Review Protocol**

Rosendo A. Rodriguez, [1] Mayra Trentin Sonoda, [2] Mohsen Agharazii, [3] Risa Shorr, [4] Kevin D. Burns [2,5]

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review **(PAGE 1, LINE 3)**
Update	1b	If the protocol is for an update of a previous systematic review, identify as such **N/A**
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number: **Page 7: lines: 1-6**
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors **(PAGE 1, LINES 4-10)**; provide physical mailing address of corresponding author **(PAGE 1, LINES: 13-18)**
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review **(PAGE 16; LINES: 1-6)**
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments **(PAGE 13 LINES 7-9)**
Support:		
Sources	5a	Indicate sources of financial or other support for the review **(PAGE 18, LINES 11-12)**
Sponsor	5b	Provide name for the review funder and/or sponsor **N/A**
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol **N/A**
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known **(PAGE5: LINES: 16-23; PAGE6: LINES:1-7)**
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) **(PAGE 6: LINES: 8-22 AND TABLE 1 -PAGE 10- see PICOS description)**
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review **(PAGE 7: LINES: 8-20; PAGE 8: LINES: 1-20)**
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage **(PAGE 9: LINES: 1-15)**

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Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated (see **SUPPLEMENTARY APPENDIX 2: PAGES 1 TO 3)**
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review **(PAGE 11, LINES 1-16);**
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility**) (PAGE 11, LINES 8-16)**; and inclusion in meta-analysis **(PAGE 12: LINES 20-22; PAGE 13 LINES 1-16)**
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators **(PAGE 11, LINES 10-20)**
Data items	12	List and define all variables for which data will be sought (as PICO items, funding sources), any pre-planned data assumptions and simplifications (**PAGE 11; LINES 20-22; PAGE 12 LINES 1-9)**
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale **(PAGE 8, LINES: 5-15)**
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis **(PAGE: 12; LINES: 10-19)**
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised **(PAGE 12: LINES 20-22; PAGE 13; LINES: 1-13)**
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) **(PAGE 13; LINES: 1-20)**
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression**) (PAGE 13: LINES: 17-20; PAGE 14; LINES 1-10)**
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned **(PAGE 13, LINES:2-5)**
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) **NOT APPLICABLE**
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) **(PAGE 14, LINES 11-14)**

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary Appendix 2. Search strategy.

Database: Embase Classic+Embase **<1947 to 2020 April 06**>, Ovid MEDLINE(R) ALL <1946 to April 06, 2020>, EBM Reviews - Cochrane Central Register of Controlled Trials <February 2020>
Search Strategy:
---
1 Nephrectomy/ (88255)
2 nephrectom*.tw,kw. (96086)
3 Kidney Transplantation/ and Living Donors/ (14605)
4 ((renal or kidney) adj3 (donor* or donation*)).tw. (32914)
5 or/1-4 (161070)
6 Vascular Stiffness/ (25452)
7 ((vascular or arter* or aort*) adj3 (stiff* or rigid*)).tw. (39922)
8 stiffness.tw,kw. (169270)
9 exp Pulse Wave Analysis/ (30543)
10 Cardio Ankle Vascular Index.tw,kw. (1710)
11 Augmentation index.tw,kw. (10033)
12 central pulse pressure.tw,kw. (1410)
13 Ankle-brachial index.tw,kw. (12896)
14 Ankle Brachial Index/ (13675)
15 (aort* adj2 (distensibilit* or elasticit*)).tw. (2677)
16 ((pulse or pulsation) adj2 (curve* or tracing* or wave*)).tw. (42852)
17 pulse wave.kw. (5300)
18 (pwv or apwv or bapwv or cfpwv).tw,kw. (20707)
19 (pulse adj2 (analys#s or velocit* or transit time)).tw. (37462)
20 (vascular stiff* or aortic stiff* or arter* stiff).kw. (1790)
21 ((decreased or reduced or diminished or lessened or lowered) adj3 ((vascular or aortic or arter*) adj compliance)).tw. (1257)
22 blood flow velocity/ (101381)
23 ((blood or circulation) adj2 (flow or rate) adj velocit*).tw. (21273)
24 (central adj (pulse or aortic or arterial) adj pressure).tw. (2881)
25 (central pressure or pulse pressure or pulse tension).tw,kw. (25275)
26 (central pulse pressure or blood flow velocit*).kw. (3437)
27 aasi.tw,kw. (560)
28 applanation tonomet*.tw,kw. (10330)
29 (sphygmocor* or vicorder*).tw,kw. (3318)
30 ((assess* or measur* or determin* or evaluat*) adj3 ((vascular or aortic or arter*) adj elasticit*)).tw. (656)
31 (Carotid adj3 intima-media thickness).tw. (23970)
32 Carotid-intima media thickness.kw. (2894)
33 Carotid artery ultrasonography.tw,kw. (292)
34 Carotid Arteries/dg (8914)
35 (Carotid arter* adj3 ultrasonograph*).tw. (1410)
36 (Ultrasonography, Doppler/ or Ultrasonography/) and Carotid Artery Diseases/ (4100)
37 calcinosis/ or exp vascular calcification/ (77074)
38 Vascular calcification.tw,kw. (11072)

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39 vascular calcinosis.tw,kw. (40)
40 Flow mediated dilatation.tw,kw. (5904)
41 Brachial Artery/ and Vasodilation/ (6138)
42 brachial artery reactivity.tw,kw. (483)
43 or/6-42 (467264)
44 5 and 43 (1078)
45 exp animals/ not exp humans/ (10091587)
46 44 not 45 (692)
47 46 use medall (302) Medline
48 46 use cctr (13) Cochrane
49 nephrectomy/ or radical nephrectomy/ (91023)
50 nephrectom*.tw. (93900)
51 kidney donor/ (11405)
52 living donor/ and kidney transplantation/ (16482)
53 ((renal or kidney) adj3 (donor* or donation*)).tw. (32914)
54 49 or 50 or 51 or 52 or 53 (165157)
55 arterial stiffness/ (27294)
56 ((vascular or arter* or aort*) adj3 (stiff* or rigid*)).tw. (39922)
57 stiffness.tw. (166817)
58 pulse wave/ (26256)
59 Cardio Ankle Vascular Index.tw. (1685)
60 augmentation index/ (5503)
61 Augmentation index.tw. (9834)
62 Augmentation index.tw. (9834)
63 pulse pressure/ (313849)
64 central pulse pressure.tw. (1389)
65 ankle brachial index/ (13675)
66 Ankle-brachial index.tw. (12479)
67 (aort* adj2 (distensibilit* or elasticit*)).tw. (2677)
68 ((pulse or pulsation) adj2 (curve* or tracing* or wave*)).tw. (42852)
69 (pwv or apwv or bapwv or cfpwv).tw. (20649)
70 (pulse adj2 (analys#s or velocit* or transit time)).tw. (37462)
71 ((decreased or reduced or diminished or lessened or lowered) adj3 ((vascular or aortic or arter*) adj compliance)).tw. (1257)
72 blood flow velocity/ (101381)
73 ((blood or circulation) adj2 (flow or rate) adj velocit*).tw. (21273)
74 (central adj (pulse or aortic or arterial) adj pressure).tw. (2881)
75 (central pressure or pulse pressure or pulse tension).tw. (24926)
76 aasi.tw. (558)
77 applanation tonomet*.tw. (10187)
78 (sphygmocor* or vicorder*).tw. (3310)
79 ((assess* or measur* or determin* or evaluat*) adj3 ((vascular or aortic or arter*) adj elasticit*)).tw. (656)
80 arterial wall thickness/ (21793)
81 (Carotid adj3 intima-media thickness).tw. (23970)
82 (Carotid arter* adj3 ultrasonograph*).tw. (1410)

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83 exp blood vessel calcification/ (19784)
84 vascular calcinosis.tw. (36)
85 Vascular calcification.tw. (10295)
86 dilatation/ and brachial artery/ (846)
87 Flow mediated dilatation.tw. (5753)
88 brachial artery reactivity.tw. (471)
89 or/55-88 (683344)
90 54 and 89 (3439)
91 exp animals/ not exp humans/ (10091587)
92 90 not 91 (1320)
93 92 use emczd (342) Embase
94 47 or 48 or 93 (657)
95 remove duplicates from 94 (524)
96 95 use medall (300) Medline
97 95 use emczd (221) Embase
98 95 use cctr (3) Cochrane

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Effects of Living Kidney Donation on Arterial Stiffness: A Systematic Review Protocol

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Effects of Living Kidney Donation on Arterial Stiffness:

A Systematic Review Protocol

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Abstract

Introduction: Kidney donors have been reported to have accelerated progression of aortic stiffness and decreased glomerular filtration compared with healthy non-donors. This is a concern because increased aortic stiffness is an independent predictor of overall cardiovascular disease and all-cause mortality in the general population. To confirm if arterial stiffness increases after donation, we will systematically review all studies that evaluated indices of arterial stiffness in healthy individuals who underwent unilateral nephrectomy for kidney donation compared to age-matched healthy non-nephrectomized controls.

Methods/Analysis: We will comprehensively search for studies published between January 1, 1960 and March 15, 2021 in MEDLINE, EMBASE, Cochrane Central, OVID, and EBM reviews. All prospective (cohort, case-control, case series, and before-and-after studies) and retrospective non-randomized studies reporting indices of arterial stiffness in nephrectomized and non-nephrectomized healthy participants will be included. Primary outcome will be the difference in the functional metrics of arterial stiffness between donors and non-donors. Secondary outcomes will be the differences in systolic/diastolic blood pressures, serum creatinine, glomerular filtration, intima-media thickness and vascular calcification. Study screening, selection and data extraction will be performed by 2 independent reviewers. Risk of bias will be independently assessed with the ROBINS-I tool and confidence in evidence by the GRADE recommendations. Qualitative and quantitative data syntheses as well as clinical and statistical heterogeneity (Forest plots, I^2 and Cochran's Q statistics) will be evaluated. If clinical and statistical heterogeneity are acceptable, inverse variance-weighted effects will be analyzed by random effect models.

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Ethics and dissemination: No ethical approval is necessary. Our results will be disseminated through peer-review publication and presentations to guide stakeholders on the evaluation and follow-up care of kidney donors.

PROSPERO registration number: CRD42020185551.

Key words: vascular, arterial, aortic stiffness, organ donation, kidney donor, cardiovascular disease, pulse wave velocity.

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Strengths and limitations of this study

- This systematic review will synthesize best quality evidence on functional and structural components of arterial stiffness in kidney donors and age-matched healthy non-nephrectomized controls.
- Since systemic blood pressure may be a determinant of arterial stiffness, its inclusion in a secondary analysis will improve our understanding on the potential relationship between blood pressure and arterial stiffness after donation.
- Quality of evidence and differences in study design can limit the strength of our associations. To minimize misinterpretations, our study will implement well-validated scales for assessing risk of bias and confidence in cumulative evidence.
- The results of this review can highlight strengths and weaknesses in previous studies and determine if there is a need for further well-designed investigations in kidney donors.

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INTRODUCTION

Live donor kidney transplantation is the preferred treatment for patients with end-stage renal disease (ESRD); it provides increased patient and graft survival and better quality of life than either dialysis or deceased donor transplantation.[1,2] Every year, nearly 20,000 healthy adults worldwide accept the risk of donor nephrectomy to help family members, friends or even strangers to improve survival and quality of living.[3,4] From a medical and ethical perspective, we ought to minimize their risks and maximize safety during and after kidney donation.

Although several studies [2,5,6] have indicated that the absolute risk for healthy kidney donors following nephrectomy is very low, some have reported that unilateral nephrectomy is associated with an accelerated progression of aortic stiffness, enhanced left ventricular mass, interventricular septum thickness and decreased glomerular filtration compared with healthy non-donors matched by age and sex.[7-11] In most of these reports, blood pressure at follow-up was not significantly different between donors and non-donors.[8,9,12,13] Moreover, the risk of developing hypertension after donation was associated with higher blood pressure readings before donation, suggesting that reduced glomerular filtration rate (GFR) post-donation may be a graded risk factor for increased aortic stiffness, independent of blood pressure.[7,13] These findings are clinically relevant because increased aortic stiffness as measured by carotid-femoral pulse wave velocity (cf-PWV) is a strong independent predictor of overall cardiovascular disease and all-cause mortality in the general population.[14] An increase of 1 m/s in cf-PWV in an otherwise healthy individual has been associated with an adjusted risk increase of 14% and 15% in total cardiovascular events and all-cause mortality.[14] Although compensatory renal hypertrophy and glomerular hyperfiltration occur after unilateral nephrectomy, a net decrease in total GFR typically follows, and this raises the possibility that cardiovascular disease risk after donation may become comparable to the risk in

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patients with mild to moderate chronic kidney disease.[15-17] A reduction in renal function of such magnitude may be particularly critical for young living kidney donors who have the longest duration of risk exposure to the effects of reduced renal mass. To determine if kidney donors are at risk for increased aortic stiffness, a systematic appraisal of existing studies on arterial stiffness in donors compared to age-matched healthy controls is necessary. This review differs from existing systematic reviews and meta-analyses that have evaluated the interventional effects of kidney transplantation on arterial stiffness in kidney recipients with ESRD.[18,19] The results of this new review can highlight strengths and weaknesses in previous studies with kidney donors and determine if there is a need for further well-designed investigations.

Objectives

In this study, we will systematically review all non-randomized studies that evaluated central or peripheral arterial stiffness in healthy individuals who underwent unilateral nephrectomy after fulfilling standard criteria for kidney donation and who had measurements of the functional or structural components of arterial stiffness, as defined by validated metrics including: carotid-femoral pulse wave velocity (PWV), carotid-radial PWV, brachial-ankle PWV, pulse wave analysis, femoral-tibial PWV, ankle-brachial index, aortic distensibility, carotid intima-media thickness and vascular calcification. The major objective of this review will be to provide best quality evidence on whether arterial stiffness increases after living kidney donation beyond what is expected for age-matched healthy non-nephrectomized controls. We will primarily determine differences in functional markers of arterial stiffness between kidney donors and their respective comparative healthy non-donor controls. Since progression of arterial stiffness in kidney donors may be a determinant of cardiovascular outcomes, we will additionally compare arterial stiffness in donors before and after nephrectomy. Secondarily, we will document differences in systolic and

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diastolic blood pressure (SBP, DBP), serum creatinine, estimated GFR (eGFR) and structural markers of arterial stiffness (carotid intima-media thickness, arterial calcification scores) between these 2 populations.

METHODS AND ANALYSIS

We will conduct this review in accordance with the Cochrane Collaboration principles for Systematic Reviews[20] and this protocol has been registered in the PROSPERO register (CRD42020185551) for systematic reviews.[21] The preferred reporting items for Systematic Reviews described by the PRISMA guidelines[22] will be followed and a checklist file for these recommendations will be included (Supplementary appendix 1).[23]

Population and eligibility criteria

Kidney donors

We will include healthy adults (≥ 18 years old) of any age who underwent unilateral nephrectomy after meeting standard screening criteria for organ donation and whose arterial stiffness was measured non-invasively before and/or after nephrectomy with any of the following functional metrics of arterial stiffness: carotid-femoral PWV, carotid-radial PWV, brachial-ankle PWV, femoral-tibial pulse PWV, ankle-brachial index, pulse wave analysis (augmentation index and central pulse pressure) and aortic distensibility.[24,25] As indicators of structural change in vascular stiffness, measurements of carotid intima-media thickness and vascular calcification scores will also be documented.[26]

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Healthy controls

As controls, we will include adult individuals (≥18 years of age) with measurements of validated indices of arterial stiffness who participated as healthy comparative controls in kidney donor studies.[7,8]

Intervention

In this review, uneventful unilateral nephrectomy for kidney donation independent of the surgical approach (laparoscopic and open nephrectomy) will be considered as the main intervention.

Outcomes

The primary outcome will be the difference in the functional metrics of arterial stiffness between healthy nephrectomized and non-nephrectomized participants as well as the difference in donors before and after nephrectomy. The absolute values in addition to between-group mean differences will be documented in their respective units of measurement. Secondary outcomes will be the differences in systolic and diastolic blood pressures (mm Hg), pulse pressure (mm Hg), serum creatinine levels (μmol/L), eGFR (ml/min/1.73 m^2), carotid intima-media thickness (mm) and vascular calcification (scores).[16,24,26,27] Clinical outcomes and any additional measures of endothelial function will be documented and reported accordingly.[17,28]

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cf-PWV reference group

In addition to the primary comparative analysis, we will search for studies that measured cf-PWV in different healthy populations worldwide and who have reported values of normal cf-PWV standardized by age.[29] We expect that this analysis will improve the generalizability of this review.

Study design

Since prospective randomized clinical trials (RCTs) of donation would never be possible for ethical reasons, only prospective non-randomized (cohort, case-control, case series, and before-and-after studies) and retrospective studies will be included, provided that 10 or more subjects have participated in the primary analysis.

Search strategy

Our search strategy will be conducted using MEDLINE, EMBASE, Cochrane Central databases, CADTH's "Grey matters light", OVID, EBM reviews, Health Technology Assessment (HTA) database and other grey literature of studies including government reports and policy statements published after January 1, 1960. A comprehensive search strategy will be constructed and implemented by a health information specialist (RS) with systematic review experience, in collaboration with the research team. MeSH terms will be used to capture the principal elements of the research question and identified citations will be exported to an excel spreadsheet for study review and selection. Our proposed literature search strategy is outlined in the supplementary appendix 2. Our planned systematic search will proceed until March 15, 2021. Manual review of

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the reference lists for all studies will be conducted according to pre-defined screening criteria (Table 1). A final "grey literature" search will be conducted using "Google Scholar" as well as reviews of *clinicaltrials.gov* for any current study. No language restriction will be utilized in any of the initial search, but our final analysis will be restricted to articles reported in English, French, Italian, Portuguese and Spanish languages. Duplicate citations will be removed and search strategies will be kept up to the time of the end of this review.

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Table 1. List of inclusion and exclusion criteria for the selection of comparative groups.

	Inclusion criteria	Exclusion criteria
Population	• Healthy subjects (≥ 18 years) who underwent unilateral simple nephrectomy as selected candidates that met standard criteria for kidney donation	• Healthy participants that underwent unilateral simple nephrectomy other than for kidney donation. • Children and adolescents with solitary kidney after unilateral nephrectomy
Intervention	• Open or laparoscopic unilateral simple nephrectomy	
Comparator	• Healthy adult subjects (≥18 years) with measurements of validated indices of arterial stiffness who participated as healthy comparative controls in kidney donor studies • Healthy subjects with measurements of carotid-femoral PWV included in reference studies	• Kidney recipients
Outcome	• Functional indices of arterial stiffness: carotid-femoral pulse wave velocity (PWV), carotid-radial PWV, brachial-ankle PWV, femoral-tibial PWV, pulse wave analysis (augmentation index and central pulse pressure), ankle-brachial index, aortic distensibility, • Structural indices of arterial stiffness: carotid intima-media thickness and arterial calcification scores.	• None
Study design	• Prospective non-randomized (cohort, case-control, case series, and before-and-after studies) and retrospective studies if 10 or more participants have been included in the primary analysis • Articles reported in English, French, Italian, Portuguese and Spanish languages	• Pediatric and non-human studies • Narrative reviews • In-vitro or mathematical modeling reports • Duplicates • Sub-studies of previously published trials. • Abstracts, conference proceedings

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Study screening, exclusions and selection

During the screening phase, we will include all prospective and retrospective non-randomized studies reporting measurements of arterial stiffness in nephrectomized and non-nephrectomized healthy participants. An iterative process of study selection will be followed using the inclusion and exclusion criteria set out in Table 1. We will exclude studies reporting on investigations in pediatric populations (age < 18 years), non-human studies, narrative reviews, in-vitro or mathematical modeling reports, abstracts or conference proceedings and any duplicate or sub-study of previously published investigations. All review processes will be independently performed by 2 individuals, with a third person available for consensus in cases of discrepancies.

All citations will first be screened by title and abstract. After selection by title/abstract and with the purpose of facilitating a final decision on eligible studies, an intermediate screening of the material and methods section will be implemented in duplicate to confirm the type of methodology employed for measuring arterial stiffness. A judicious decisive selection before data extraction will be achieved by the independent reviewers at the full-text level. In cases of missing information, we will consider contacting study authors. Neither of the review authors will be blinded to the journal titles or to the study authors or institutions.

Data extraction

A data extraction form will be prepared a-priori with consensus from all investigators and optimized using a subset of 4 randomly selected studies prior to duplicate extraction by 2 independent reviewers. When multiple publications arise from one study, relevant data will be extracted into a single form. Data extraction will include: a) *study characteristics, design and methods*: title, authors, journal/source/year, language of publication, country, type of study design,

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study period, publication status, total number of donors and non-donors, inclusion and exclusion criteria, and points of measurement; b) *sample characteristics*: age, sex, age at time of nephrectomy, age at time of assessment, duration of follow-up and type of arterial stiffness instrumentation; c) *outcomes*: change in arterial stiffness based on the metric of measurement: carotid-femoral PWV, carotid-radial PWV, brachial-ankle PWV, femoral-tibial pulse PWV, ankle-brachial index, augmentation index, central pulse pressure, aortic distensibility, carotid artery intima-media thickness; arterial calcification scores, serum creatinine levels, eGFR, systolic, diastolic and pulse blood pressure. We will document if arterial stiffness values are reported as adjusted or non-adjusted according to changes in mean arterial blood pressure and/or heart rate. If reported, we will document type and number of medications for the control of blood pressure.

Risk assessment of bias

Risk of bias for non-randomized studies will be assessed using the ROBINS-I tool.[30] Seven domains through which bias might be introduced will be independently evaluated and discussed by 2 reviewers, encompassing confounding, selection of participants (nephrectomized and non-nephrectomized), classification of the intervention (i.e. unilateral nephrectomy), deviation from the intended intervention, missing data, measurement of outcomes and selection of the reported results. Each domain will be judged as either low, moderate, serious or critical risk of bias or no information available. After discussion among reviewers, an overall assessment of study bias with inclusion of all domains will be tabulated. Unresolved disagreements will be resolved with participation of a third reviewer.

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Data synthesis and analysis plan

Studies will be included in qualitative and quantitative syntheses if they fulfill all eligibility criteria. Study characteristics will be summarized using means and standard deviations or median and inter-quartile ranges for continuous variables and numbers and percentages for categorical variables. A narrative report of study characteristics will also be provided. We will identify potential sources of clinical heterogeneity according to differences in study design characteristics, methodological quality, characteristics at baseline between nephrectomized groups and their controls, and duration of follow-up periods. If at least 2 studies report on the same outcome, a quantitative analysis (i.e. meta-analysis) will be attempted on those studies. Statistical heterogeneity will be characterized with Forest plots, the I^2 and Cochran's Q statistics ($p<0.10$). If both clinical and statistical heterogeneity are acceptable, inverse variance-weighted effects meta-analysis will be performed. An I^2 value less than 60% with a non-significant chi-square statistic ($P > 0.10$) will be suggestive of moderate statistical heterogeneity. We will primarily choose the random effects model according to the methodology of DerSimonian and Laird,[31] but a fixed-effects meta-analysis will also be modeled as part of our sensitivity analysis. Based on the scale of measurements, we may choose to calculate pooled effect estimates using either standardized or mean differences and their corresponding 95% confidence intervals (95% CI). To minimize the "double counting" error in before-and-after studies, half of the total number of study participants will be allocated to each study arm. All analysis will be performed using RevMan 5.3 (The Nordic Cochrane Centre, The Cochrane Collaboration, 2014, Copenhagen).

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Sensitivity analysis

Depending on the number of eligible studies, sensitivity analyses may be performed to evaluate the effects of extreme values, high risk of study bias, and any confounding effects associated with differences in the reporting of adjusted or non-adjusted values of arterial stiffness according to changes in mean arterial pressure and/or heart rate. An analysis of the differences between fixed and random effects models on the pooled effect estimates will also be assessed.

Subgroup analysis

Clinical studies[32] have documented that living donors genetically related to their ESRD recipients are more likely to develop the primary kidney disease. Thus, it is critical to differentiate the effects of unilateral nephrectomy on arterial stiffness between recipient-related healthy controls and those who are not related to the recipient. To minimize bias in our review and depending on the number of eligible studies, nature and detail of the reported information, we will attempt to determine the effects of potential confounders by performing sub-group analyses based on age, surgical technique (open vs laparoscopic nephrectomy), duration of follow-up after nephrectomy (short vs long-term), type of comparator (recipient-related versus non-recipient-related), number and type of medications used for the control of blood pressure and sex according to different male to female ratios. Inter-group differences will be analyzed using the Cochran's Q statistics with $p \leq 0.10$.

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Confidence in cumulative evidence

To assess the certainty in the evidence and strength of recommendations on the effects of unilateral nephrectomy in kidney donors, 2 reviewers will evaluate quality of evidence for each outcome measure according to the 5 domains of GRADE recommendations.[33]

Amendments

Any protocol amendments will be summarized in the form of a Table, where date of amendment, description of changes and rationale will be provided.

Patient and public involvement

Patients and/or the public were not involved in the design of the protocol.

DISCUSSION

Because living kidney donors are selected amongst the healthiest individuals, the long-term risk of ESRD, cardiovascular disease or death has been reported to be similar to or even lower than that in the general population.[6] It has been documented however, that unilateral nephrectomy frequently leads to reduced kidney function and variable levels of increased serum creatinine.[5,15,16] Although the medium- and long-term adverse effects of hyperfiltration and compensatory hypertrophy after donation remain unclear, endothelial dysfunction associated with decreased brachial artery reactivity[17,34] and increased levels of uremic toxins[35] have been reported in kidney donors. More recently, some studies have reported that unilateral nephrectomy leads to an increase in aortic stiffness, left ventricle mass and decreased GFR as early as one year after surgery.[7-9] Although blood pressure was not markedly affected in these studies, initial results suggest that

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reduced GFR may be an independent graded risk factor for increased aortic stiffness and adverse left ventricle remodeling, independent of blood pressure.[8,11,12]

To characterize the hemodynamic effects of unilateral nephrectomy in kidney donors, we propose to systematically identify, appraise and summarize all non-randomized studies that have investigated indices of arterial stiffness before and after nephrectomy in donors to determine any increase in the progression of arterial stiffness compared with age-matched healthy individuals. In addition to improving our understanding of the relationship between reduced kidney function after donation and cardiovascular function, our findings will determine if unilateral nephrectomy is a risk factor for increased arterial stiffness in donors and whether non-invasive metrics of arterial stiffness could be useful for risk stratification.

Animal studies have shown that alterations in renal development resulting in reduced nephron number in the offspring result in blood pressure elevation, reduced cardiac functional reserve and left ventricle hypertrophy in adulthood.[36] Although alterations of the renin-angiotensin-aldosterone system and changes in vascular tone are frequently present after unilateral nephrectomy,[37] kidney donation is not considered a risk factor for hypertension post-donation. Kasiske et al[38] followed living kidney donors for a 3-year period and identified an increase in blood pressure over time, but differences between donors and controls were non-significant. Kim et al[39] found that there was an increase in blood pressure after donation with 21% of donors requiring medication for control of hypertension. Thiel et al[13] reported that the risk of developing hypertension (defined as >140/90 mm Hg) increased 3.6-fold 1-year post-donation, but baseline values of kidney donors who became hypertensive were significantly higher than those who were less susceptible to hypertension. Because systemic blood pressure may be a determinant of arterial stiffness, we will include systolic and diastolic blood pressure as a secondary outcome. This will

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add to the understanding of the potential relationship between blood pressure and arterial stiffness following organ donation.[37-39]

To accept that increased arterial stiffness is associated with the effects of unilateral nephrectomy, convincing evidence based on good quality studies and adequate methodological designs is critical. Although quality of evidence and differences in study design are factors that may limit the strength of our associations, implementing well-validated scales in our study for the assessment of risk of bias and certainty of evidence will minimize misinterpretations.[30,33] In cases where information is not available, we will make every possible effort to contact the primary author(s) to obtain any missing information.

Living kidney donors obtain no medical benefit from donation, exposing themselves to health risks associated with the surgical procedure and the effects of reduced renal mass for the benefit of another human being. Promotion of donor autonomy in the medical decision-making process requires accurate quantification of the mid and long-term risks after donation. This is particularly critical for young living donors who have the longest risk of exposure.[39,40] Our review will provide a more accurate estimate of the "short-to-mid-term" risks of cardiovascular disease after donation by examining the progression of arterial stiffness post-nephrectomy.[37-41] Such results can inform potential donors and may guide their follow-up care and preserve good health in the long-term.

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Ethics and dissemination

No ethics approval is required for this systematic review. Our findings will be disseminated through peer-review publication and presentation, which can inform clinical guidelines and management of potential kidney donors.

Contributors: RAR and KDB conceived and designed the study; RAR created the analytical plan and drafted the protocol; RAR and RS designed the search strategy; RAR, MTS, MA, and KDB will be involved in one or more of the following tasks: study screening and selection, data extraction, verification, quality appraisal, synthesis, analysis of the evidence and data interpretation. All authors have read, reviewed and approved the final version of the manuscript. RAR is the guarantor.

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Competing interest: None declared

Patient and consent for publication: Not required

Provenance and peer-review: Not commissioned; externally peer-reviewed.

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Supplementary Appendix 1: PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

TITLE: **Effects of Living Kidney Donation on Arterial Stiffness: A Systematic Review Protocol**

Rosendo A. Rodriguez, [1] Mayra Trentin Sonoda, [2] Mohsen Agharazii, [3] Risa Shorr, [4] Kevin D. Burns [2,5]

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review, **(PAGE 1, LINE 3)**
Update	1b	If the protocol is for an update of a previous systematic review, identify as such **N/A**
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number: Page 3 line 4; Page 7: lines: 6-7
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors **(PAGE 1, LINES 4-10**); provide physical mailing address of corresponding author **(PAGE 1, LINES: 13-19)**
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review **(PAGE 19; LINES: 5-10)**
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments **(PAGE 16 LINES 5-7)**
Support:		
Sources	5a	Indicate sources of financial or other support for the review **(PAGE 19, LINES 11-12)**
Sponsor	5b	Provide name for the review funder and/or sponsor **N/A**
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol **N/A**
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known **(PAGE5: LINES: 16-23; PAGE 6: LINES:1-7)**
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) **(PAGE 6: LINES: 10-23; PAGE 7, LINES 1-3).**
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review**. (page 8, lines 1-20; PAGE 9: LINES: 12-21); and TABLE 1 -PAGE 11- (see PICOS description)**
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other

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		grey literature sources) with planned dates of coverage (**PAGE 9: LINES: 1-15; PAGE 10, LINES 1-6))**
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated (see **SUPPLEMENTARY APPENDIX 2: PAGES 1 TO 3)**
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review **(PAGE 12, LINES 10-16);**
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility**) (PAGE 12, LINES 1-16)**; and inclusion in meta-analysis **(PAGE 14: LINES 5-13;**
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators **(PAGE 12, LINES 17-22)**
Data items	12	List and define all variables for which data will be sought (as PICO items, funding sources), any pre-planned data assumptions and simplifications (**PAGE 12; LINES 20-22; PAGE 13 LINES 1-10)**
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale **(PAGE 8, LINES: 11-19)**
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis **(PAGE: 13; LINES: 12-20)**
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised **(PAGE 14: LINES 1-5)**
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) **(PAGE 14; LINES: 5-20)**
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression**) (PAGE 15: LINES: 1-6; PAGE 15; LINES 8-19)**
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned **(PAGE 14, LINES:1-5)**
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) **NOT APPLICABLE**
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) **(PAGE 16, LINES 1-4)**

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary Appendix 2. Search strategy.

Database: Embase Classic+Embase **<1947 to 2020 April 06**>, Ovid MEDLINE(R) ALL <1946 to April 06, 2020>, EBM Reviews - Cochrane Central Register of Controlled Trials <February 2020>
Search Strategy:
---
1 Nephrectomy/ (88255)
2 nephrectom*.tw,kw. (96086)
3 Kidney Transplantation/ and Living Donors/ (14605)
4 ((renal or kidney) adj3 (donor* or donation*)).tw. (32914)
5 or/1-4 (161070)
6 Vascular Stiffness/ (25452)
7 ((vascular or arter* or aort*) adj3 (stiff* or rigid*)).tw. (39922)
8 stiffness.tw,kw. (169270)
9 exp Pulse Wave Analysis/ (30543)
10 Cardio Ankle Vascular Index.tw,kw. (1710)
11 Augmentation index.tw,kw. (10033)
12 central pulse pressure.tw,kw. (1410)
13 Ankle-brachial index.tw,kw. (12896)
14 Ankle Brachial Index/ (13675)
15 (aort* adj2 (distensibilit* or elasticit*)).tw. (2677)
16 ((pulse or pulsation) adj2 (curve* or tracing* or wave*)).tw. (42852)
17 pulse wave.kw. (5300)
18 (pwv or apwv or bapwv or cfpwv).tw,kw. (20707)
19 (pulse adj2 (analys#s or velocit* or transit time)).tw. (37462)
20 (vascular stiff* or aortic stiff* or arter* stiff).kw. (1790)
21 ((decreased or reduced or diminished or lessened or lowered) adj3 ((vascular or aortic or arter*) adj compliance)).tw. (1257)
22 blood flow velocity/ (101381)
23 ((blood or circulation) adj2 (flow or rate) adj velocit*).tw. (21273)
24 (central adj (pulse or aortic or arterial) adj pressure).tw. (2881)
25 (central pressure or pulse pressure or pulse tension).tw,kw. (25275)
26 (central pulse pressure or blood flow velocit*).kw. (3437)
27 aasi.tw,kw. (560)
28 applanation tonomet*.tw,kw. (10330)
29 (sphygmocor* or vicorder*).tw,kw. (3318)
30 ((assess* or measur* or determin* or evaluat*) adj3 ((vascular or aortic or arter*) adj elasticit*)).tw. (656)
31 (Carotid adj3 intima-media thickness).tw. (23970)
32 Carotid-intima media thickness.kw. (2894)
33 Carotid artery ultrasonography.tw,kw. (292)
34 Carotid Arteries/dg (8914)
35 (Carotid arter* adj3 ultrasonograph*).tw. (1410)
36 (Ultrasonography, Doppler/ or Ultrasonography/) and Carotid Artery Diseases/ (4100)
37 calcinosis/ or exp vascular calcification/ (77074)
38 Vascular calcification.tw,kw. (11072)

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39 vascular calcinosis.tw,kw. (40)
40 Flow mediated dilatation.tw,kw. (5904)
41 Brachial Artery/ and Vasodilation/ (6138)
42 brachial artery reactivity.tw,kw. (483)
43 or/6-42 (467264)
44 5 and 43 (1078)
45 exp animals/ not exp humans/ (10091587)
46 44 not 45 (692)
47 46 use medall (302) Medline
48 46 use cctr (13) Cochrane
49 nephrectomy/ or radical nephrectomy/ (91023)
50 nephrectom*.tw. (93900)
51 kidney donor/ (11405)
52 living donor/ and kidney transplantation/ (16482)
53 ((renal or kidney) adj3 (donor* or donation*)).tw. (32914)
54 49 or 50 or 51 or 52 or 53 (165157)
55 arterial stiffness/ (27294)
56 ((vascular or arter* or aort*) adj3 (stiff* or rigid*)).tw. (39922)
57 stiffness.tw. (166817)
58 pulse wave/ (26256)
59 Cardio Ankle Vascular Index.tw. (1685)
60 augmentation index/ (5503)
61 Augmentation index.tw. (9834)
62 Augmentation index.tw. (9834)
63 pulse pressure/ (313849)
64 central pulse pressure.tw. (1389)
65 ankle brachial index/ (13675)
66 Ankle-brachial index.tw. (12479)
67 (aort* adj2 (distensibilit* or elasticit*)).tw. (2677)
68 ((pulse or pulsation) adj2 (curve* or tracing* or wave*)).tw. (42852)
69 (pwv or apwv or bapwv or cfpwv).tw. (20649)
70 (pulse adj2 (analys#s or velocit* or transit time)).tw. (37462)
71 ((decreased or reduced or diminished or lessened or lowered) adj3 ((vascular or aortic or arter*) adj compliance)).tw. (1257)
72 blood flow velocity/ (101381)
73 ((blood or circulation) adj2 (flow or rate) adj velocit*).tw. (21273)
74 (central adj (pulse or aortic or arterial) adj pressure).tw. (2881)
75 (central pressure or pulse pressure or pulse tension).tw. (24926)
76 aasi.tw. (558)
77 applanation tonomet*.tw. (10187)
78 (sphygmocor* or vicorder*).tw. (3310)
79 ((assess* or measur* or determin* or evaluat*) adj3 ((vascular or aortic or arter*) adj elasticit*)).tw. (656)
80 arterial wall thickness/ (21793)
81 (Carotid adj3 intima-media thickness).tw. (23970)
82 (Carotid arter* adj3 ultrasonograph*).tw. (1410)

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83 exp blood vessel calcification/ (19784)
84 vascular calcinosis.tw. (36)
85 Vascular calcification.tw. (10295)
86 dilatation/ and brachial artery/ (846)
87 Flow mediated dilatation.tw. (5753)
88 brachial artery reactivity.tw. (471)
89 or/55-88 (683344)
90 54 and 89 (3439)
91 exp animals/ not exp humans/ (10091587)
92 90 not 91 (1320)
93 92 use emczd (342) Embase
94 47 or 48 or 93 (657)
95 remove duplicates from 94 (524)
96 95 use medall (300) Medline
97 95 use emczd (221) Embase
98 95 use cctr (3) Cochrane

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